Joint Labor Search and the Tax-Transfer System

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Motivation

Motivation I: joint labor search

- Starting point: families coordinate their labor supply.

- In frictional labor markets: families choose a *joint job-search strategy*.
  - how hard to search, on and off the job (e.g., endogenous search intensity);
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$\Rightarrow$ This model is a natural laboratory to study tax-and-transfer and UI policies. Two-person households have multiple private insurance channels on top of public insurance.

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We ask: does the unit of taxation or the generosity of UI matter more for couples' employment and welfare; and what would be the optimal combination of UI and tax policy? Separate taxation but joint UI?

Contribution

→ First equilibrium search model with households' joint job search, precautionary savings, *and* endogenous job creation, applied to study joint tax-transfer policies.

Couples insure themselves against job-loss risk through three private channels that interact in general equilibrium:

1. within-household risk sharing (added-worker effect);
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We run three sets of policy experiments:

1. joint → separate progressive taxation;
2. UI reforms: generosity, duration, and *spouse-dependent* eligibility.
3. We look for the optimal combination of UI and tax policy.

▸ related literature

Preview of results

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For couples, the unit of the tax-transfer system matters more than its generosity. The status quo — pooling income for *taxation* but individualizing *UI* — may have the wrong unit on each instrument.

Model

Households

- Continuous time, stationary equilibrium.
- Mass μ couples (two workers) and $1-\mu$ singles (we focus on couples).
- Workers: (i) employed (E), (ii) non-employed (N; $N = \{U, I\}$).
- Households: (i) dual-earner (EE), (ii) single earner (EN), (iii) no earner (NN).
- Risk averse; the flow utility of a couple is

$$2\,u(c/2) - k(s_1) - k(s_2),$$

  where consumption is shared equally; $k(\cdot)$ is a convex search cost function.

- Rate of time preference ρ.
- Perpetual youth: retire at rate λ, no bequests. Reborn as NN, no assets.
- Save in a risk-free asset at rate r; no borrowing ($a \geq 0$).

$\rightarrow$ Self-insurance is the only savings motive.

Firms, matching, separations

- Firms post vacancies at flow cost K; free entry, taking into account composition of searchers (household type, assets)
- Match productivity $y \sim H$, fixed for the match.
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- Vacancy fills at $q(\theta) = p(\theta)/\theta$.
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- Separations of an employed worker:
  - layoff at rate δ;
  - reallocation shock at rate $\chi p(\theta)$: new draw $y' \sim H$, accept or fall into N;
  - quit opportunity at rate γ (logit choice with taste shocks, scale ξ);
  - job-to-job move on an accepted outside offer.

Wages and government

Wages. Bargaining protocol of Elsby–Gottfries, yields a simple linear sharing rule, independent of the worker's outside option.

$$w(y) = \eta\, y.$$

Government. Household disposable income (HSV / Bénabou):

$$N^c(Y) = \max\left[\tau_0^c \, Y^{1-\tau_1^c},\ \bar{f}^c\right], \qquad N^s(Y) = \max\left[\tau_0^s \, Y^{1-\tau_1^s},\ \bar{f}^s\right].$$

- τ_1: progressivity; $\bar{f}$: income floor (food stamps, vouchers).
- UI benefits $b^x(y)$ depend on the last wage and (optionally) the spouse's employment status $x \in \{n, e\}$; UI expires at rate ζ; benefits are taxed (enter Y).
- Balanced budget; G fixed in counterfactuals, τ_0^c adjusts.

Household Bellman equation (dual earners, ee)

▸ single-earner v^{en} ▸ no-earner v^{nn}

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\begin{aligned}
(\rho+\lambda+2\delta+2\gamma)\, v^{ee}(y_1,y_2,a) = \max_{c,s_1,s_2} \; & 2u(c/2)-k(s_1)-k(s_2) \\
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- **Savings** (line 6): asset drift – self-insurance.

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dual-earner job J^{ee}

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- **Assets drift** – self-insurance.

Calibration and Model Fit

Data

- Survey of Income and Program Participation (SIPP), 1996–2005 waves.
- Allows us to observe assets (we use liquid) and the reason for leaving the last job (quits).

- Sample: prime-age (25–55), stable couples (spouse present, no turnover) and never married, labor market attached; drop self-employed, disabled, students, retirees, perm. inactive.
- **55,096** individuals; **1.33M** person-months; **18,092** couples.

- We use regressions to calculate our calibration targets, taking into account survey design and compositional differences (e.g., age, education, gender, children),

Couples	Dual-earner (EE)	Single-earner (EN)	No-earner (NN)
Share	72.6%	26.5%	0.9%

Calibration

Externally set ($\mu = 1$: couples only in the baseline):

$\sigma = 2$ (CRRA)	$r = 0.0017$ (interest)	$\lambda = 0.0021$ (retirement)	$\eta = 0.5$
$\psi = 0.4$ (repl. rate)	$\zeta = 0.1823$ (6m UI)	$\bar{x} = 3.22$ (Faberman et al.)	$\gamma = 1$
$(\tau_0^c, \tau_1^c) = (0.785, 0.133)$ – Krueger–Wu, joint schedule			

Internally calibrated (μ_y normalizes average income to 1) to match

Moment	Target	Param	Moment	Target	Param
Median assets/income	3.46	ρ	EE rate	1.66%	φ
NE rate	7.10%	K	EN rate	1.01%	δ
EE w/ wage cut	43.2%	χ	Mean-min wage ratio	1.67	σ_y^2
Quit share (EN)	20.7%	ξ	Food-stamp floor	10%	$\bar{f}^c$

ρ: time preference; φ: search-cost curvature; K: vacancy cost; δ: layoff rate; χ: reallocation rate; μ_y, σ_y^2: productivity distribution; $\bar{f}^c$: income floor; ξ: quit-shock scale.

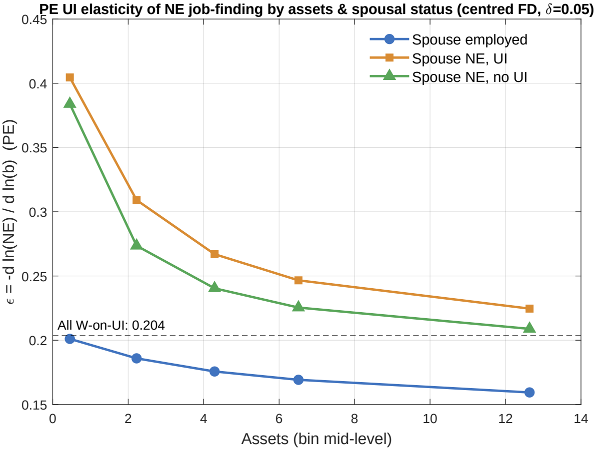
Model fit: stocks and flows

	Data	Model		Data	Model
Employment	87.5	86.4	NE rate	7.10†	7.10
Nonemployment	12.5	13.6	Zero earners	16.50	15.69
Single-earner couples	26.5	25.2	One earner	5.11	6.42
Dual-earner couples	72.6	73.8			
No-earner couples	0.9	1.0			
EE rate	1.66†	1.64	EN rate	1.01†	0.99
One earner	1.60	1.97	One earner	0.90	0.82
Dual, primary	1.48	1.27	Dual, primary	0.60	0.89
Dual, secondary	1.59	1.92	Dual, secondary	1.30	1.13
EE rate ($\Delta w < 0$)	0.71	0.72	EN rate (vol. quit)	0.20†	0.21
One earner	0.71	0.80	One earner	0.18	0.06
Dual, primary	0.73	0.91	Dual, primary	0.09	0.09
Dual, secondary	0.60	0.50	Dual, secondary	0.29	0.36

† Calibration target. All numbers in percent; disaggregated (by earners) moments are untargeted.

▸ distributions (plots) ▸ wage & asset moments

An external test: the job-finding elasticity



- Average elasticity of the job-finding rate w.r.t. UI $\approx$ 0.20; up to 0.26 for liquidity-constrained couples.
- Falls with household assets; highest when the spouse *also* receives UI.
- Within the empirically supported range (Le Barbanchon–Schmieder–Weber, 2024).

▸ consumption drop on job loss

Policy Experiments

Welfare

- Veil of ignorance for a newly entering household (zero assets, *NN*).
- By how much, x, would we have to scale *everyone's* consumption in the **benchmark** economy – at its own steady-state policy – to make this *newborn* as well off as under the reforms?
- We fix quit policies and search effort, and scale consumption of all households from benchmark c^* to $(1+x)\,c^*$.
- We then recompute all value functions in the benchmark economy and solve implicitly for the x that makes the newborn indifferent between benchmark and counterfactual.

⇒ $x > 0$ means the reform raises welfare; being ex-ante, the criterion rewards insurance against job-loss and earnings risk.

- Caveat: an ex-ante *newborn*, *steady-state* comparison – it abstracts from transition dynamics and redistribution among currently-living households.

Policy experiments: roadmap

Hold UI fixed, vary tax system:

1. Separate taxation (joint $\rightarrow$ separate progressive, budget-balanced): employment $+1.6$ pp via secondary earners; welfare $+0.09\%$. ▸ table

Hold tax system fixed, vary UI:

1. UI generosity ($\psi : 0.4 \rightarrow 0.5$, joint tax): tiny effects; welfare -0.05% – crowds out precautionary saving. ▸ table
2. Decomposition of the UI welfare effect (joint tax): the direct insurance gain is undone by the saving response. ▸ table
3. Optimal UI level (joint tax): welfare-maximizing flat $\psi^* \approx 0.30 \approx$ current – couples already well-insured. ▸ figure
4. Spouse-dependent UI (joint tax): more generous when the spouse is non-employed; welfare $+0.09\%$. ▸ figure

$\Rightarrow$ **What is the optimal combination?** separate progressive taxes + spouse-dependent UI *(next)*.

Optimal combination: separate progressive taxes + spouse-dependent UI

		Δ Welfare (% cons.)	
τ_1^C	Optimal (ψ_E, ψ_N)	Optimal SD-UI	Flat ψ=0.40
0.100	(0.20, 0.50)	−0.273	−0.346
0.115	(0.20, 0.50)	−0.066	−0.144
0.130	(0.20, 0.50)	+0.139	+0.055
0.145	(0.20, 0.50)	+0.342	+0.252
0.160	(0.20, 0.50)	+0.539	+0.445
0.175	(0.20, 0.50)	+0.735	+0.636
0.190	(0.20, 0.50)	+0.928	+0.823
0.205	(0.10, 0.50)	+1.120	+1.007
0.220	(0.10, 0.50)	+1.310	+1.188
0.235	(0.10, 0.50)	+1.496	+1.365

Budget rebalanced at each node; free entry clears. "SD-UI" = optimum over the (ψ_E, ψ_N) grid; "Flat" = benchmark UI. Shaded: benchmark progressivity $\tau_1^C = 0.133$. ▸ employment effects

Optimal combination: discussion

- Most of the welfare gain comes from higher progressivity under separate filing: the flat-UI column rises from $+0.06\%$ at the benchmark ($\tau_1^C \approx 0.13$) to $+1.37\%$ at 0.235. This is the social-insurance value of progressive taxation (HSV, Krueger-Wu).
- New: Spouse-dependent UI adds a roughly constant ≈ 0.1 pp *on top* at every progressivity level – the same gain as the headline spouse-dependence experiment.
- At benchmark progressivity the reforms are *modest*; the larger gains also require *also* raising progressivity.
- Optimal UI is more generous when the spouse is non-employed ($\psi_N = 0.50$); under separate progressive taxation this exceeds the joint-tax-baseline optimum ($\psi_N = 0.40$). When the spouse is employed, the replacement rate is between 0.1 and 0.2 (ψ_E).
- Employment is nearly flat in progressivity: $+1.6$ pp at the benchmark (the secondary-earner effect of separate filing), rising only to $+1.9$ pp at $\tau_1^C = 0.235$ – the large welfare gains are *insurance*, not jobs. SD-UI adds $\approx +0.1$ pp throughout.
- No decomposition yet (work in progress).

Conclusions

Conclusions

- An equilibrium joint-search model where couples self-insure through three channels – within-household risk sharing, savings, and job search – interacting with endogenous job creation.
- Separate taxation raises employment (≈ 1.6 pp) by drawing in secondary earners; wages and inequality barely move; welfare gain modest.
- UI generosity is near-optimal: private channels make extra public insurance largely redundant. The one improvement is spouse-dependent UI.
- Optimal combination: separate *progressive* taxation *plus* spouse-dependent UI delivers the largest welfare gains – growing with progressivity, with spouse-dependence adding ≈ 0.1 pp on top.

For couples, the unit of the tax-transfer system matters more than its generosity: tax individually, condition UI on the spouse.

Thank you

Backup: single-earner couple (v^{en})

$$
\begin{aligned}
(\rho+\lambda+\delta+\gamma)\, v^{en}(y_1, y_2, a) = \max_{c, s_1, s_2}\ & 2u(c/2) - k(s_1) - k(s_2) \\
& + \bar{x} s_1 p(\theta) \int \max\left[v^{en}(y_1', y_2, a) - v^{en}, 0\right] dH(y_1') \\
& + \chi p(\theta) \int \left[\max\{v^{en}(y_1', y_2, a), v^{nn}\} - v^{en}\right] dH(y_1') \\
& + s_2 p(\theta) \int \max\left[v^{ee}(y_1, y_2', a) - v^{en}, 0\right] dH(y_2') \\
& + \delta\, v^{nn} + \tfrac{\gamma}{\xi} \ln\left[e^{\xi v^{nn}} + e^{\xi v^{en}}\right] \\
& + \zeta\left[v^{en}(y_1, 0, a) - v^{en}\right] \\
& + \tfrac{dv^{en}}{da}\left[ra - c + N^c(w(y_1) + b^e(y_2))\right].
\end{aligned}
$$

◂ back to v^{ee}

Backup: no-earner couple (v^{nn})

$$
\begin{aligned}
(\rho+\lambda)\, v^{nn}(y_1, y_2, a) = \max_{c, s_1, s_2}\; & 2u(c/2) - k(s_1) - k(s_2) \\
& + s_1 p(\theta) \int \max\left[v^{en}(y_1', y_2, a) - v^{nn}, 0\right] dH(y_1') \\
& + s_2 p(\theta) \int \max\left[v^{ne}(y_1, y_2', a) - v^{nn}, 0\right] dH(y_2') \\
& + \zeta\left[v^{nn}(0, y_2, a) - v^{nn}\right] + \zeta\left[v^{nn}(y_1, 0, a) - v^{nn}\right] \\
& + \frac{dv^{nn}}{da}\left[ra - c + N^c(b^n(y_1) + b^n(y_2))\right].
\end{aligned}
$$

back to v^{ee}

Backup: dual-earner job value (J_1^{ee})

Job filled by worker 1 in a dual-earner couple:

$$
\begin{aligned}
\left[\rho^f + \lambda + \delta + \chi p(\theta) + S_1^{ee}\bar{x}p(\theta)\bar{H}(y_1) + \gamma\pi_1^{ee}\right] J_1^{ee}(y_1, y_2, a) &= y_1 - w(y_1) \\
&+ \bar{x}S_2^{ee}\, p(\theta)\int_{y_2}\left[J_1^{ee}(y_1, y_2', a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&+ \chi p(\theta)\int\left[\mathbf{1}_{\{y_2'\geq R^e\}} J_1^{ee}(y_1, y_2', a) + \mathbf{1}_{\{y_2'<R^e\}} J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&+ \left[\delta + \gamma\pi_2^{ee}\right]\left[J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] \\
&+ \frac{dJ_1^{ee}}{da}\left[ra - c + N^c(w(y_1) + w(y_2))\right].
\end{aligned}
$$

No simultaneous quit when the spouse finds work – quits arise only via the γ opportunity.

back to J^{en}

Backup: dual-earner job value (J_1^{ee})

Job filled by worker 1 in a dual-earner couple:

$$
\begin{aligned}
&\left[\rho^f + \lambda + \delta + \chi p(\theta) + S_1^{ee}\bar{x}p(\theta)\bar{H}(y_1) + \gamma\pi_1^{ee}\right] J_1^{ee}(y_1, y_2, a) = y_1 - w(y_1) \\
&\quad + \bar{x}S_2^{ee}\, p(\theta)\int_{y_2}\left[J_1^{ee}(y_1, y_2', a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \chi p(\theta)\int\left[\mathbf{1}_{\{y_2' \geq R^e\}} J_1^{ee}(y_1, y_2', a) + \mathbf{1}_{\{y_2' < R^e\}} J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \left[\delta + \gamma\pi_2^{ee}\right]\left[J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] \\
&\quad + \frac{dJ_1^{ee}}{da}\left[ra - c + N^c(w(y_1) + w(y_2))\right].
\end{aligned}
$$

- **Worker 1's job ends** (LHS): retirement, layoff, reallocation, J2J move, or quit (rate γ only).

No simultaneous quit when the spouse finds work – quits arise only via the γ opportunity.

back to J^{en}

Backup: dual-earner job value (J_1^{ee})

Job filled by worker 1 in a dual-earner couple:

$$
\begin{aligned}
&\left[\rho^f + \lambda + \delta + \chi p(\theta) + S_1^{ee}\bar{x}p(\theta)\bar{H}(y_1) + \gamma\pi_1^{ee}\right] J_1^{ee}(y_1, y_2, a) = y_1 - w(y_1) \\
&\quad + \bar{x}S_2^{ee}\, p(\theta)\int_{y_2}\left[J_1^{ee}(y_1, y_2', a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \chi p(\theta)\int\left[\mathbf{1}_{\{y_2' \geq R^e\}} J_1^{ee}(y_1, y_2', a) + \mathbf{1}_{\{y_2' < R^e\}} J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \left[\delta + \gamma\pi_2^{ee}\right]\left[J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] \\
&\quad + \frac{dJ_1^{ee}}{da}\left[ra - c + N^c(w(y_1) + w(y_2))\right].
\end{aligned}
$$

- **Worker 1's job ends** (LHS): retirement, layoff, reallocation, J2J move, or quit (rate γ only).
- **Spouse climbs the ladder**: a job-to-job move to a better job.

No simultaneous quit when the spouse finds work – quits arise only via the γ opportunity.

back to J^{en}

Backup: dual-earner job value (J_1^{ee})

Job filled by worker 1 in a dual-earner couple:

$$
\begin{aligned}
&\left[\rho^f + \lambda + \delta + \chi p(\theta) + S_1^{ee}\bar{x}p(\theta)\bar{H}(y_1) + \gamma\pi_1^{ee}\right] J_1^{ee}(y_1, y_2, a) = y_1 - w(y_1) \\
&\quad + \bar{x}S_2^{ee}\, p(\theta) \int_{y_2} \left[J_1^{ee}(y_1, y_2', a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \chi p(\theta) \int \left[\mathbf{1}_{\{y_2' \geq R^e\}} J_1^{ee}(y_1, y_2', a) + \mathbf{1}_{\{y_2' < R^e\}} J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \left[\delta + \gamma\pi_2^{ee}\right]\left[J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] \\
&\quad + \frac{dJ_1^{ee}}{da}\left[\, ra - c + N^c(w(y_1) + w(y_2))\,\right].
\end{aligned}
$$

- **Worker 1's job ends** (LHS): retirement, layoff, reallocation, J2J move, or quit (rate γ only).
- **Spouse climbs the ladder**: a job-to-job move to a better job.
- **Spouse's job is disrupted**: reallocation redraw (stay *ee* or drop to *en*) or separation (layoff/quit $\to$ *en*).

No simultaneous quit when the spouse finds work – quits arise only via the γ opportunity.

back to J^{en}

Backup: dual-earner job value (J_1^{ee})

Job filled by worker 1 in a dual-earner couple:

$$
\begin{aligned}
&\left[\rho^f + \lambda + \delta + \chi p(\theta) + S_1^{ee}\bar{x}p(\theta)\bar{H}(y_1) + \gamma\pi_1^{ee}\right] J_1^{ee}(y_1, y_2, a) = y_1 - w(y_1) \\
&\quad + \bar{x}S_2^{ee}\, p(\theta) \int_{y_2} \left[J_1^{ee}(y_1, y_2', a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \chi p(\theta) \int \left[\mathbf{1}_{\{y_2' \geq R^e\}} J_1^{ee}(y_1, y_2', a) + \mathbf{1}_{\{y_2' < R^e\}} J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] dH(y_2') \\
&\quad + \left[\delta + \gamma\pi_2^{ee}\right]\left[J^{en}(y_1, y_2, a) - J_1^{ee}(y_1, y_2, a)\right] \\
&\quad + \frac{dJ_1^{ee}}{da}\left[ra - c + N^c(w(y_1) + w(y_2))\right].
\end{aligned}
$$

- **Worker 1's job ends** (LHS): retirement, layoff, reallocation, J2J move, or quit (rate γ only).
- **Spouse climbs the ladder**: a job-to-job move to a better job.
- **Spouse's job is disrupted**: reallocation redraw (stay *ee* or drop to *en*) or separation (layoff/quit $\to$ *en*).
- **Assets drift** – self-insurance.

No simultaneous quit when the spouse finds work – quits arise only via the γ opportunity.

back to J^{en}

Related literature

Joint vs. separate taxation Guner–Kaygusuz–Ventura (2012); Bick–Fuchs-Schündeln (2018); Borella–De Nardi–Yang (2023); Holter–Krueger–Stepanchuk (2023).
– *frictionless labor markets.*

Joint decisions in search models Guler–Guvenen–Violante (2012); Flabbi–Mabli (2018); Pilossoph–Wee (2021).
– *partial equilibrium.*

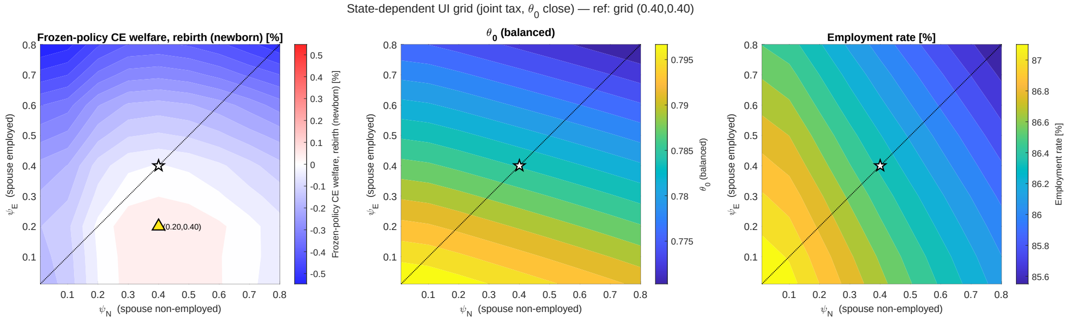
Secondary earners + endogenous labor demand Mankart–Oikonomou (2017); Birinci (2021).
– *no on-the-job search.*

Precautionary savings + job search Clymo–Denderski–Mercan–Shoefer (2026); Chaumont–Shi (2022); Aiyagari (1994); Ortigueira–Siassi (2013).

Unemployment insurance Le Barbanchon–Schmieder–Weber (2024); Koehne–Kuhn (2015).

full references back

Spouse-dependent UI: full grid



Left: frozen-policy CE welfare; middle: budget-balancing tax level θ_0; right: employment rate. $\star$ = flat benchmark $(0.40, 0.40)$, $\triangle$ = optimum.

back

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Le Barbanchon, T., J. F. Schmieder, and A. Weber (2024). "Job search, unemployment insurance, and active labor market policies," *Handbook of Labor Economics*, vol. 5.

Mankart, J. and R. Oikonomou (2017). "Household search and the aggregate labour market," *Review of Economic Studies*.

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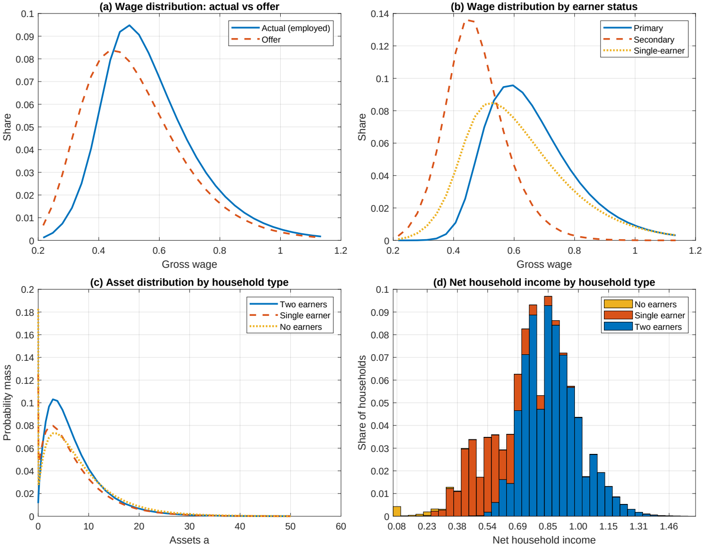
Backup: model fit – distributions



Wages, household income and assets: the model reproduces the residualized wage distribution and substantial wealth inequality across couples.

◂ back to fit

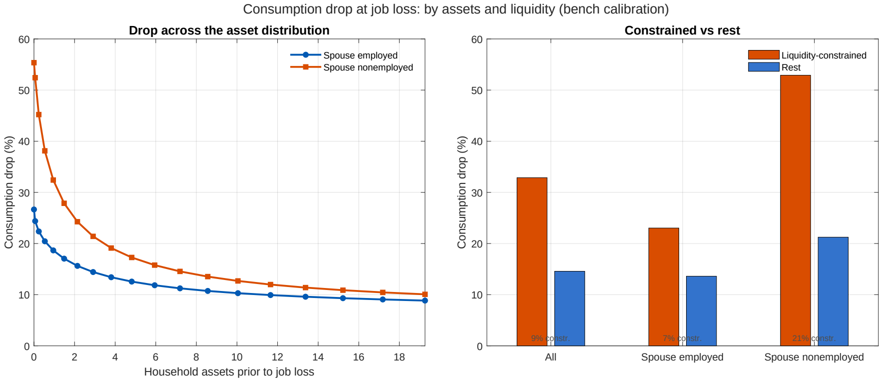
Backup: model fit – wage & asset distributions

	Data	Model		Data	Model
A. Wages			*C. Assets*		
P10†	0.73	0.69	P10†	0.00	0.07
P50†	0.99	0.96	P50†	1.00	0.73
P90†	1.28	1.37	P90†	1.92	2.28
P50/P10	1.35	1.39	P50/P10	70.54	16.00
P90/P50	1.29	1.43	P90/P50	1.92	2.64
Mean/Min(P5)	1.67	1.61			
B. Relative wages			*D. Assets/Income*		
EN couple, empl	1.01	1.07	Median†‡	3.46	3.47
EE couple, primary	1.05	1.15	P10†	−0.92	0.42
EE couple, secondary	0.96	0.83	P90†	7.26	10.39

† Relative to respective mean. ‡ Calibration target.

back to fit

Backup: consumption drop upon job loss



- Average drop ≈ 16% (gross income falls 60%): ≈ 14% if the spouse is employed, ≈ 30% if non-employed, and near one-to-one (≈ 52%) when the household is also liquidity-constrained.
- Within-household insurance and precautionary savings are quantitatively large.

back

Experiment 1: joint $\rightarrow$ separate taxation

Balanced-budget reform; progressivity τ_1^c fixed, level τ_0^c adjusts.

	Baseline (joint)	Separate filing
Employment	86.4	88.0
Dual-earner couples	73.8	77.0
Single-earner couples	25.2	22.1
$N \rightarrow E$ rate, one-earner couples	6.42	7.39
Mean wage / inequality	0.57	0.57 (unch.)
Δ Welfare (CE)	–	+0.09%

Lower marginal rates on secondary earners $\Rightarrow$ they enter employment. Effect is on participation, not the wage distribution. Welfare gain positive but modest.

roadmap

Experiment 2: more generous UI ($\psi : 0.4 \to 0.5$)

	Baseline	$\psi = 0.5$
Employment	86.4	86.2
Average assets	4.77	4.64
Δ Cons. upon job loss (%)	16.3	15.5
Δ Welfare (CE)	–	-0.05%

- Small effects: more generous public insurance crowds out precautionary saving.

roadmap

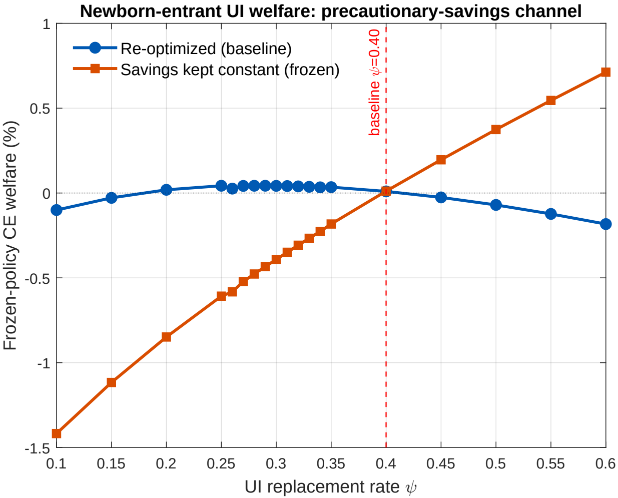
Decomposition of the welfare effect ($\psi : 0.4 \to 0.5$)

	Δ Welfare (%)	Δ Employment (pp)
Direct (insurance, no response)	$+0.32$	-0.23
$+$ Consumption–savings response	-0.29	$+0.13$
$+$ Labor demand (vacancies)	-0.08	-0.01
$+$ Quits / search / interactions	$+0.00$	-0.12
Total	$\mathbf{-0.05}$	$\mathbf{-0.23}$

The direct insurance gain ($+0.32$) is almost entirely undone once couples adjust precautionary saving (-0.29); labor demand adds a further negative effect.

roadmap

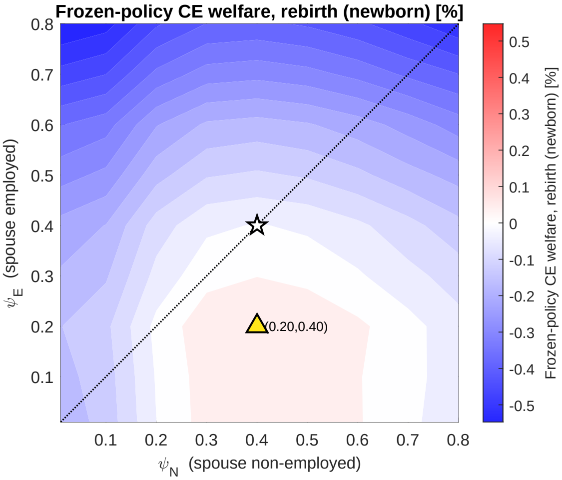
Optimal level of UI



- Welfare-maximizing $\psi \approx 0.30$ – close to (slightly below) the current 0.40.
- Couples are, if anything, *slightly over-insured*.
- Red line: with savings frozen, cutting UI would hurt – the result hinges on the consumption-saving margin.

roadmap

Spouse-dependent UI



Welfare over (ψ_N, ψ_E); $\star$ benchmark, $\triangle$ optimum.

roadmap

- Let ψ_E, ψ_N depend on the spouse's employment status.
- Optimal: $\psi_E = 0.20$, $\psi_N = 0.40$ – more generous when the spouse is non-employed.
- Public insurance is most valuable exactly when within-household insurance is absent.
- Welfare gain $\approx +0.09\%$.

θ_0 & employment

Optimal combination: employment effects

τ_1^C	Optimal (ψ_E, ψ_N)	Δ Employment (pp)	
		Optimal SD-UI	Flat ψ=0.40
0.100	(0.20, 0.50)	+1.59	+1.50
0.115	(0.20, 0.50)	+1.64	+1.55
0.130	(0.20, 0.50)	+1.70	+1.60
0.145	(0.20, 0.50)	+1.75	+1.65
0.160	(0.20, 0.50)	+1.80	+1.70
0.175	(0.20, 0.50)	+1.86	+1.74
0.190	(0.20, 0.50)	+1.91	+1.79
0.205	(0.10, 0.50)	+2.08	+1.84
0.220	(0.10, 0.50)	+2.13	+1.88
0.235	(0.10, 0.50)	+2.19	+1.93

Change in employment (pp) vs the joint-progressive benchmark (86.4%); tax level rebalances the budget, free entry clears. Shaded row: benchmark progressivity.

back